TECHNICAL DESIGN & ARCHITECTURE SPECIFICATION

Profile & Message Screen UI Redesign

A comprehensive mobile redesign for Linkhain featuring Seller Sold Portfolios with Made-to-Order Custom Commissions, Buyer Saved Favorites, and Bi-Directional Campus Chat with Pinned Escrow Terms.

TARGET PROJECT	SCREENS REDESIGNED	DELIVERY FORMAT	AIA SOURCE INTEGRITY
Linkhain Campus Hub	**UserProfile & Inbox**	**Interactive HTML + PDF**	**Preserved (No AIA overwrite)**

1. Executive Summary & Design Rationale

This document details the refined user interface architecture for Linkhain's two most critical transaction touchpoints: the **User Profile Screen (`UserProfile`)** and the **Message / Negotiation Screen (`Inbox`)**. In university peer-to-peer marketplaces, two recurring challenges degrade trust and repeat commerce:

Seller Inactivity After Stock Depletion: Student creators and artisans (e.g., crochet artists, digital template creators, 3D/resin makers) sell single unique items that quickly disappear from active feeds, leaving their profiles empty and preventing interested buyers from ordering replicas.

Unstructured Chat Negotiations: Campus meetup coordination via generic chat feeds often results in missed rendezvous, unclear price confirmations, and confusion over who spoke last.

> **Core Innovation:** The redesigned **Profile Screen** introduces a dedicated **Seller Portfolio of Past Sold Items** with an active **`[ ✦ Commission Similar Item ]`** CTA button, paired with a **Buyer Wishlist of Saved Favorites**. The redesigned **Message Screen** introduces **Bi-Directional Speech Bubbles** (Seller on the left in light card style; Buyer on the right in signature Maroon) anchored by a **Sticky Pinned Deal Card** and **Instant Campus Quick-Reply Chips**.

2. Visual System & Design Tokens

Design Token	Hex Value	Usage & Semantic Purpose
Deep University Maroon	#5F1C2E	Primary brand color, top app bars, buyer speech bubbles, primary CTAs.
University Gold	#CCA668 / #A8843C	Verified seller accents, star ratings, commission sparkles, badge borders.

Design Token	Hex Value	Usage & Semantic Purpose
Warm Paper	`#F7F4F1`	Mobile canvas background, card backing, subtle separators.
Card White	`#FFFFFF`	Floating cards, seller speech bubbles, modal sheets, product cards.
Verified Teal	`#0D9488` / `#10B981`	Institutional student verification badges, live campus presence indicators.
Sold Crimson	`#991B1B`	Distinct sold-out badges on completed portfolio items.

3. Side-by-Side UI Architecture Comparison

The mockup below displays both redesigned screens side-by-side as rendered in modern mobile viewports (390×844 dp), showcasing their shared visual harmony, layout containment, and color system.

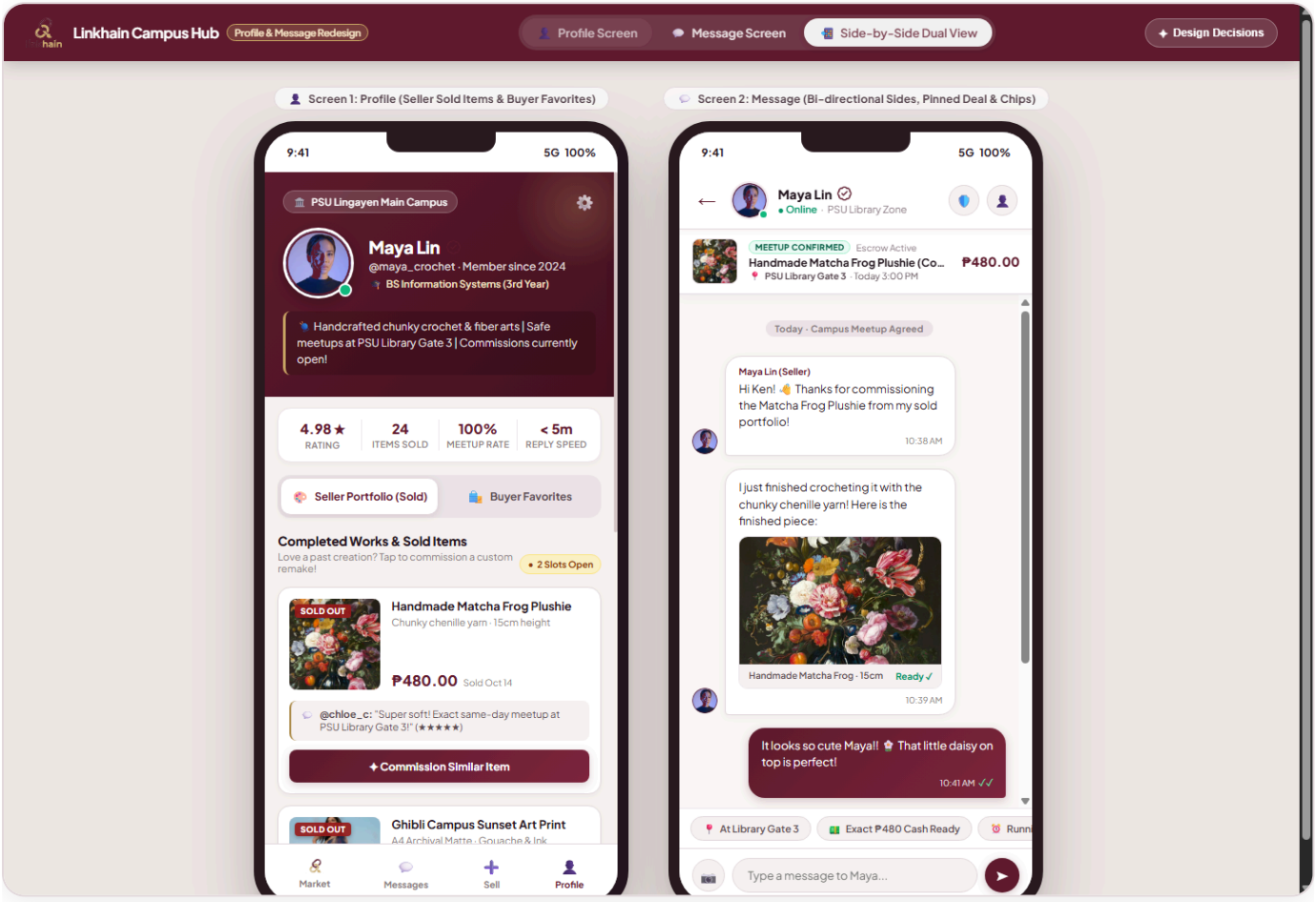
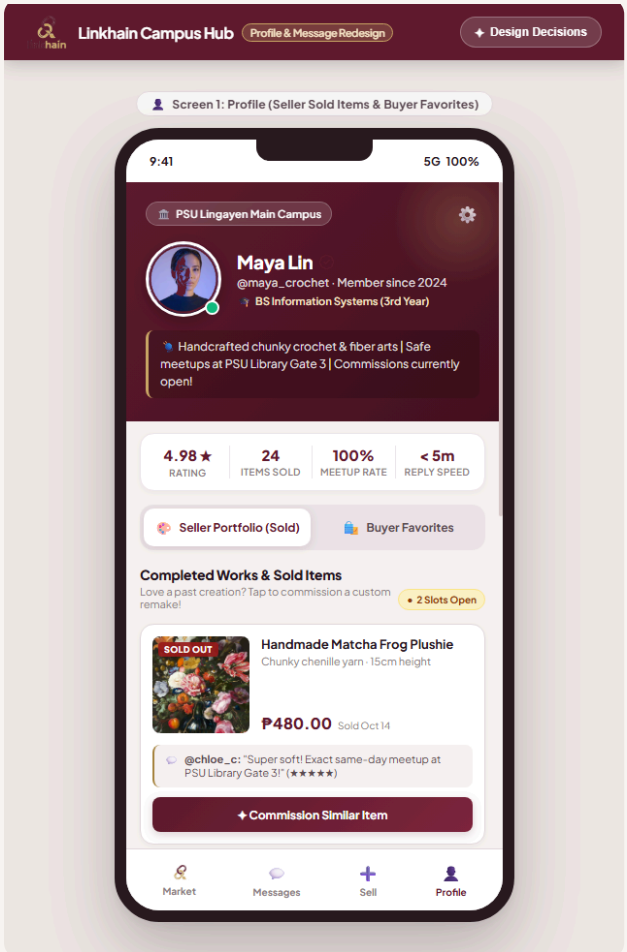


Figure 1: Side-by-Side UI Prototype — Screen 1: Profile (Left) and Screen 2: Message (Right)

Responsive Containment: Both screens adhere to strict 90% mobile container rules, eliminating the text clipping and component cut-off issues previously observed in raw MIT App Inventor layouts.

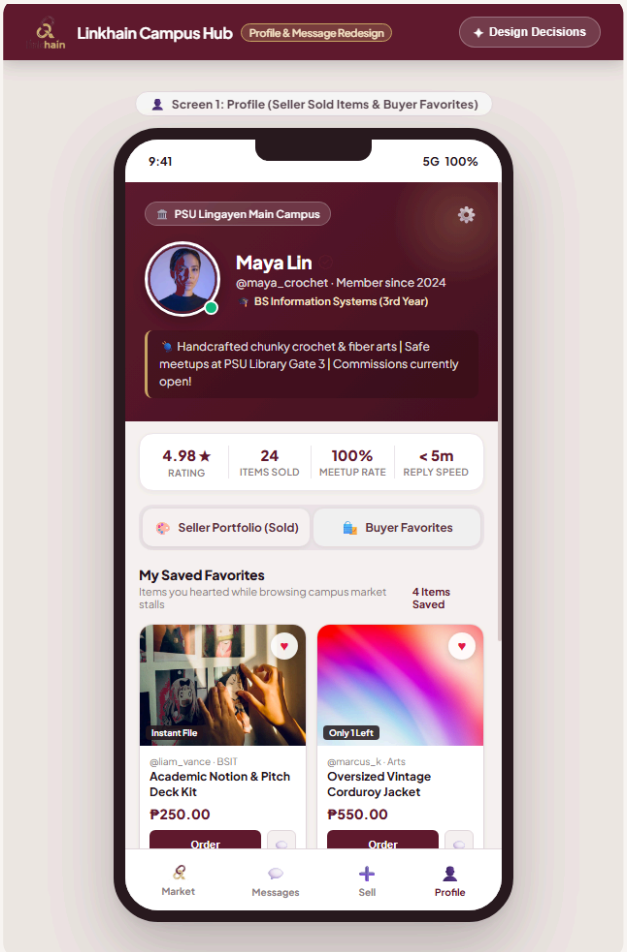
4. Profile Screen Walkthrough (`UserProfile`)

The redesigned Profile screen incorporates an interactive segmented role switcher allowing students to seamlessly toggle between **Seller Portfolio** and **Buyer Favorites**.



A. Seller Mode: Sold Portfolio

Displays past sold creations with "SOLD OUT" badges and the "✦ Commission Similar Item" CTA.



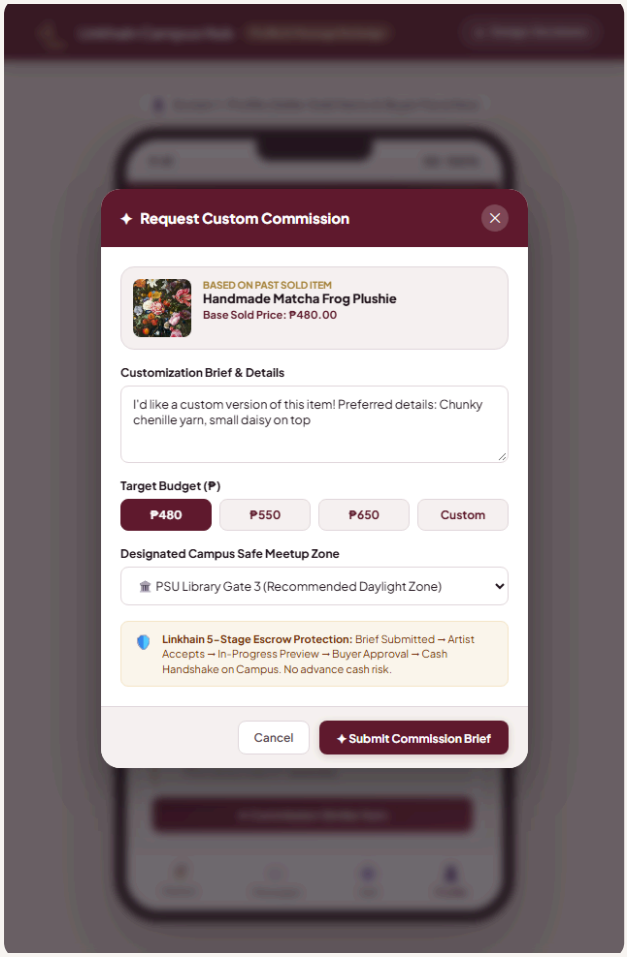
B. Buyer Mode: Saved Favorites

Displays hearted marketplace items with active red hearts, prices, and direct meetup order buttons.

Profile Component	Interactive Behavior	Technical Value
Institutional Hero Card	Displays avatar, name, verified checkmark, department (BSIS 3B), and campus location.	Builds immediate peer trust and eliminates campus identity fraud.
4-Metric Trust Bar	Aggregates ratings (4.98 ★), total items sold (24), meetup rate (100%), and response time (<5m).	Presents student seller reliability at a glance.
Sold Portfolio Grid	Lists completed student works with historical sold price (₱480) and verified student reviews.	Turns past sales into marketing assets for custom re-orders.

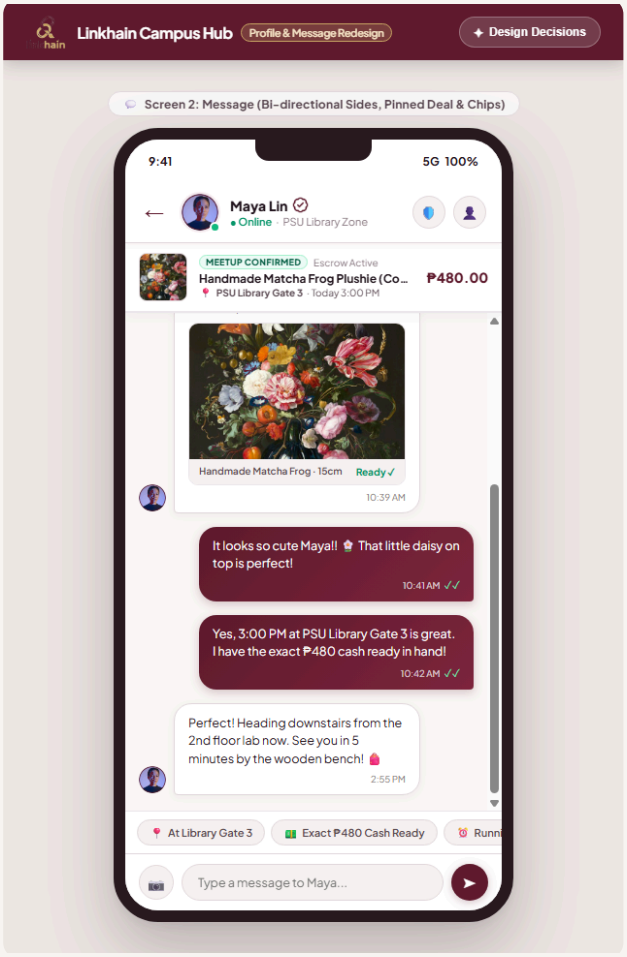
Profile Component	Interactive Behavior	Technical Value
Saved Favorites Grid	2-column grid of saved cards with live stock tags and active heart indicators.	Provides a direct personal wishlist for campus buyers.

5. Interactive Custom Commission Sheet & Message Screen



C. Commission Request Sheet

Pre-populates item info, customization notes, budget chips, and daylight safety zone picker.



D. Bi-Directional Message Screen

Pinned deal context card, Left/Right bubble alignment, and instant-tap campus chips.

6. Message Screen Key Innovations (Inbox)

Bi-Directional Side Alignment: Incoming messages from the student seller appear on the **Left** with their verified avatar and white card backing; outgoing buyer messages appear on the **Right** in signature Maroon with teal delivery ticks (✓✓ Read).

Sticky Pinned Transaction Card: The item being negotiated (e.g. *Handmade Matcha Frog Plushie · ₱480.00 · PSU Library Gate 3*) stays permanently anchored at the top of the chat, eliminating price ambiguity during meetup handoffs.

One-Tap Campus Quick Replies: Horizontal scroll chips allow students to tap preset campus status updates ("*At Library Gate 3*", "*Exact ₱480 Cash Ready*", "*Running 5m Late*") with automated realistic seller replies.

7. Cloud Database Architecture & Data Schema

The redesigned Profile and Message workflows interface with the Linkhain Redis-backed MIT CloudDB service using structured key-value serialization:

Data Domain	CloudDB Key Format	Schema & Payload Description
User Accounts	`user:<email>`	Stores student name, campus ID, department, verification status, and credentials.
Seller Sold Items	`sold:<seller_email>`	JSON list of completed items: title, original sold price (₱), photos, and buyer reviews.
Buyer Favorites	`favs:<buyer_email>`	JSON list of saved listing IDs bookmarked by the student buyer.
Commission Contracts	`comm:<comm_id>`	5-stage milestone tracking: Brief Submitted → Artist Accepts → In-Progress Preview → Buyer Approved → Completed.
Chat Threads	`thread:<thread_id>`	Ordered list of messages: `[sender_email, message_text, timestamp, media_url, is_read]`.

8. Public Project Deliverables & Access for Groupmates

To allow groupmates, instructors, and reviewers to interact with the prototype directly without installation, the complete package is published online:

Online Interactive Prototype:

`https://cutiekyee.github.io/linkhain-ui/`

Anyone with this link can test the interactive Profile toggle, commission modal, and live chat composer directly in any mobile or desktop web browser.

Project Source & Deliverables Archive:

`https://github.com/cutiekyee/linkhain-ui`

Contains the standalone HTML prototype, printable PDF documentation, high-resolution annotated screenshots, and MIT App Inventor design tokens.

Deliverable File	Format	Location / Verification
Interactive UI Prototype	Standalone HTML5 (Self-Contained Base64)	`Linkhain_Profile_Message_Redesign.html`
Comprehensive PDF Report	Print-Ready PDF Document	`Linkhain_Profile_Message_Redesign_Report.pdf`
Original Source AIA	MIT App Inventor 2 (.aia)	`Linkhain.aia` (Untouched & preserved)
Compiled Android Package	Android Application Package (.apk)	`Linkhain_V15.apk`